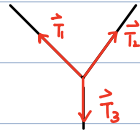
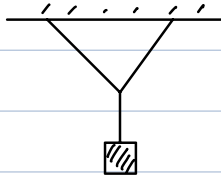


• Equilibrium

1° $\vec{F}_{net} = 0$



$$\vec{T}_1 + \vec{T}_2 + \vec{T}_3 = 0$$



$$-\vec{T}_3 + m\vec{g} = 0$$

2° $\vec{\tau}_{net} = 0$

measured from ANY point

- { Net force = 0 → translational equilibrium
- { Net torque = 0 → rotational equilibrium

Equilibrium:

$$\begin{cases} \vec{p} = \text{constant} & (\text{momentum}) \\ \vec{L} = \text{constant} & (\text{angular momentum}) \end{cases}$$

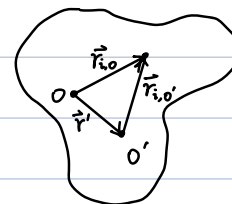
Static equilibrium:

$$\begin{cases} \vec{p} = 0 \\ \vec{L} = 0 \end{cases}$$

When $\begin{cases} \sum_i \vec{F}_i = 0 \\ \sum_i \vec{\tau}_{i,o} = 0 \end{cases} \Rightarrow \sum_i \vec{\tau}_{i,o'} = 0 \text{ for any other point } o'$

Proof:

$$\begin{aligned} \sum_i \vec{\tau}_{i,o} &= \sum_i \vec{r}_{i,o} \times \vec{F}_i \\ &= \sum_i (\vec{r}' + \vec{r}_{i,o'}) \times \vec{F}_i \\ &= \vec{r}' \times (\sum_i \vec{F}_i) + \sum_i \vec{r}_{i,o'} \times \vec{F}_i \\ &= \sum_i \vec{\tau}_{i,o'} \end{aligned}$$



①

Center of gravity (cog) :

Gravitational force $\vec{F}_g$ effectively acts on a single point.

cog = COM under uniform gravity.

Proof :

$$\begin{cases} \vec{\tau}_{net} = \sum_i \vec{r}_i \times m_i \vec{g}_i \\ \vec{\tau}_{eff} = \vec{r}_{cg} \times (\sum_i m_i \vec{g}_i) \end{cases}$$

for uniform gravity

$$\vec{g}_i = \vec{g}$$

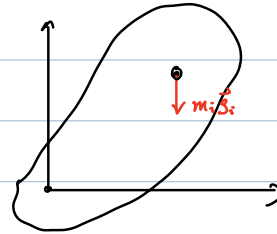
$$\Rightarrow \begin{cases} \vec{\tau}_{net} = \sum_i \vec{r}_i \times m_i \vec{g} \\ \vec{\tau}_{eff} = \vec{r}_{cg} \times (M \vec{g}) \end{cases}$$

$$\text{if } \vec{r}_{cg} = \vec{r}_{com} = \frac{\sum_i m_i \vec{r}_i}{M}$$

$$\Rightarrow \vec{\tau}_{eff} = \frac{1}{M} (\sum_i m_i \vec{r}_i) \times (M \vec{g})$$

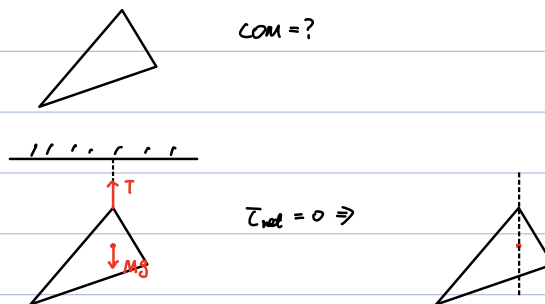
$$= \sum_i m_i \vec{r}_i \times \vec{g}$$

$$= \vec{\tau}_{net}$$

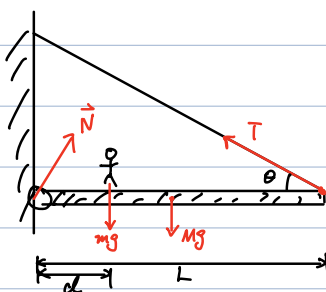


In other words, we can put a single force $M\vec{g}$ on the point cog to calculate $\vec{\tau}_{net}$

Example.



Example : Standing on a beam



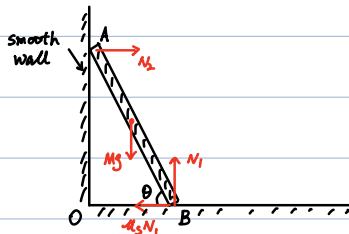
$$mgd + Mg \frac{L}{2} = TL \sin \theta$$

$$\Rightarrow T = (mgd + Mg \frac{L}{2}) / (L \sin \theta)$$

$$\begin{cases} N_x = T \cos \theta \\ N_y = mg + Mg - T \sin \theta \end{cases}$$

(2)

Example: A leaning ladder



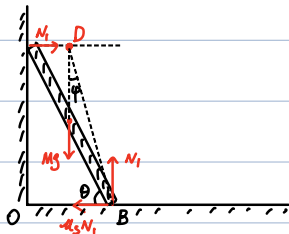
Q: minimal θ for static equilibrium?

Method -1. (torque around A)

$$\begin{cases} N_1 = Mg \\ Mg \frac{L}{2} \cos \theta + \mu_s N_1 L \sin \theta = N_1 L \cos \theta \end{cases}$$

$$\Rightarrow \mu_s Mg L \sin \theta = (Mg L - Mg \frac{L}{2}) \cos \theta$$

$$\Rightarrow \tan \theta = \frac{1}{2\mu_s}$$



Method-2 (torque around D = 0)

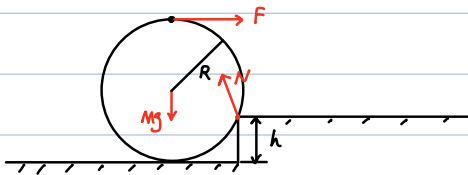
$$\Rightarrow \frac{\mu_s N_1}{N_1} = \tan \varphi$$

$$\tan \varphi = \frac{1}{2} \cot \theta$$

$$\Rightarrow \mu_s = \frac{1}{2} \cot \theta$$

$$\Rightarrow \tan \theta = \frac{1}{2\mu_s}$$

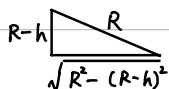
Example.



Q1: Minimal F?

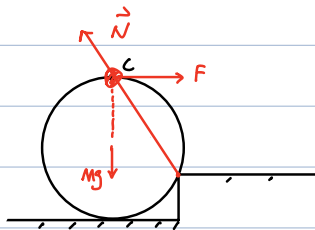
$$F(2R - h) = Mg \sqrt{R^2 - (R - h)^2}$$

$$\Rightarrow F = Mg \frac{\sqrt{R^2 - (R - h)^2}}{2R - h}$$



Q2: direction of $\vec{N}$?

$$\vec{\tau}_{\text{net}, C} = 0 \Rightarrow$$



• Simple Harmonic motion

Stable equilibrium:

$$\begin{cases} F = -\frac{dU}{dx} = 0 & (\text{equilibrium}) \\ \frac{d^2U}{dx^2} > 0 & (\text{stable}) \end{cases}$$

i.e. minimum of $U(x)$

e.g. Harmonic oscillator

$$U(x) = \frac{1}{2} k x^2$$

$$\left. \begin{aligned} \frac{dU}{dx} &= kx = 0 \Rightarrow x=0 \\ \frac{d^2U}{dx^2} &= k > 0 \end{aligned} \right\} \Rightarrow x=0 \text{ stable equilibrium}$$

Taylor expansion:

$$U(x) = U(x_0) + \underbrace{\frac{dU}{dx}\bigg|_{x_0}}_{\substack{\downarrow \\ \text{equilibrium position}}} (x-x_0) + \frac{1}{2} \underbrace{\frac{d^2U}{dx^2}\bigg|_{x_0}}_k (x-x_0)^2 + \dots$$

$$\text{for } x \sim x_0, \quad U(x) \approx U(x_0) + \frac{1}{2} k (x-x_0)^2$$

This universal family of motion: Simple Harmonic Motion

$$\text{Force: } F = -\frac{dU}{dx} = -k(x-x_0)$$

$$\Rightarrow a = \frac{F}{m} = -\frac{k}{m}(x-x_0)$$

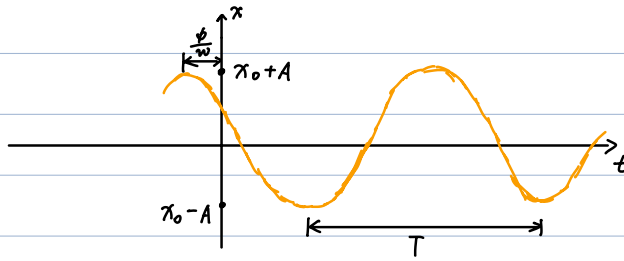
i.e.,

$$\begin{aligned} \ddot{x} &= -\frac{k}{m}(x-x_0) \\ \Rightarrow \left\{ \begin{array}{l} \boxed{x-x_0 = A \cos(\omega t + \phi)} \\ \omega \equiv \sqrt{\frac{k}{m}} \end{array} \right. & \text{Angular frequency} \end{aligned}$$

A : amplitude

$T \equiv \frac{2\pi}{\omega}$: period

ϕ : phase constant / phase angle



$$t = -\frac{\phi}{\omega} \Rightarrow x - x_0 = A$$

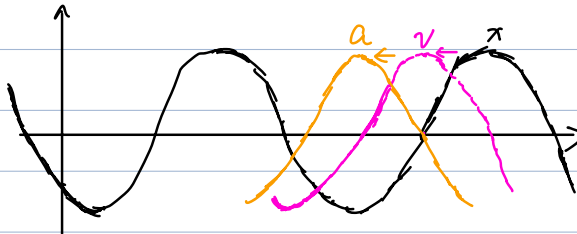
frequency: (cycles per second):

$$f \equiv \frac{1}{T} = \frac{\omega}{2\pi}$$

unit: s^{-1} , or hertz (Hz)

$$x - x_0 = A \cos(\omega t + \phi)$$

$$\Rightarrow \begin{cases} v = -\omega A \sin(\omega t + \phi) = \omega A \cos(\omega t + \phi + \frac{\pi}{2}) \\ a = -\omega^2 A \cos(\omega t + \phi) = \omega^2 A \cos(\omega t + \phi + \pi) \end{cases}$$



Note:

$$1^\circ \quad a = -\omega^2 (x - x_0)$$

proportional to $(x - x_0)$, opposite sign

2° $\{x, v, a\}$ same ω , different phases

Energy:

$$K = \frac{1}{2} m v^2 = \frac{1}{2} m \omega^2 A^2 \sin^2(\omega t + \phi)$$

$$= \frac{1}{2} k A^2 \sin^2(\omega t + \phi)$$

$$\omega = \sqrt{\frac{k}{m}}$$

$$U = \frac{1}{2} k (x - x_0)^2 = \frac{1}{2} k A^2 \cos^2(\omega t + \phi)$$

$$E = K + U = \frac{1}{2} k A^2$$

Note:

$$K = \frac{1}{2} k A^2 \frac{1 - \cos(2\omega t + 2\phi)}{2}$$

$$U = \frac{1}{2} k A^2 \frac{1 + \cos(2\omega t + 2\phi)}{2}$$

frequency: 2ω

Energy transformation:

